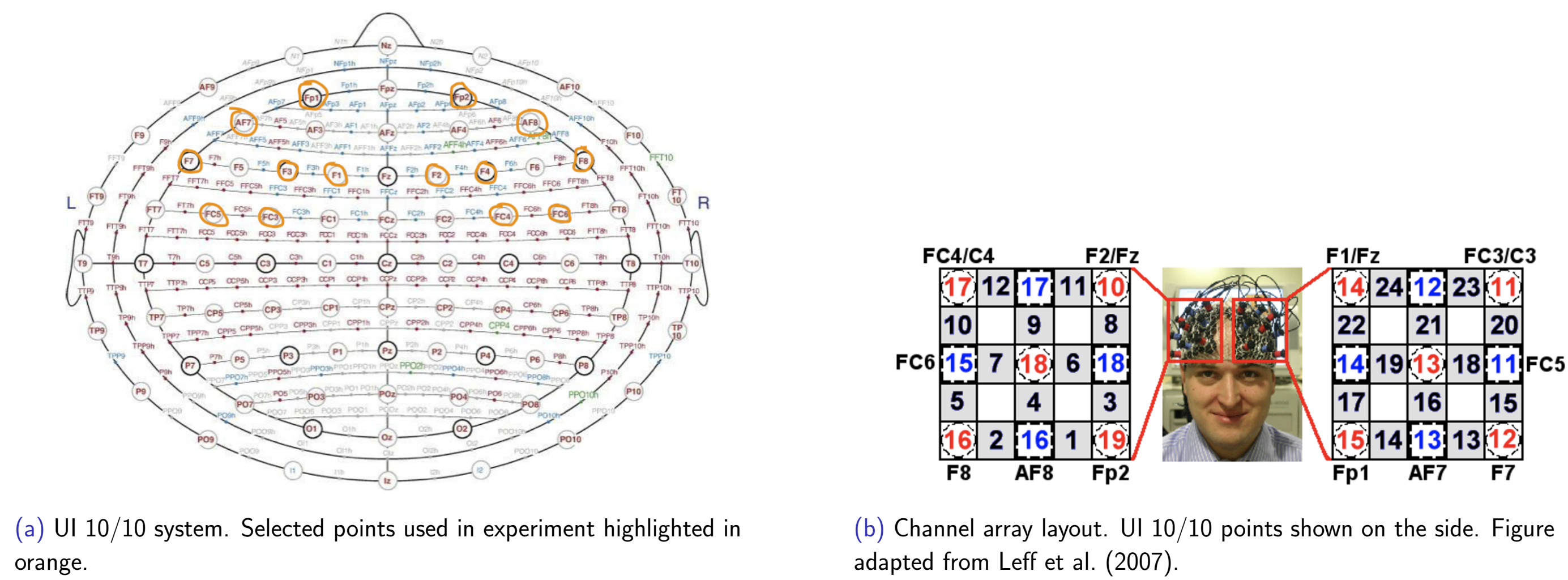


fNIRS surgical reef knot tying hypothesis testing

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This project uses the fNIRS database of brain activity of novice, registrar and consultant surgeons in a reef knot tying task collected by Leff et al. (2007, 2008). An algorithm extracts 8 scalar values for each subject from the raw fNIRS signal. It is provided by Dr Felipe Orihuela-Espina and Daniel Leff and uses the ICNNA library (Orihuela-Espina et al., 2017). The contributions of this work are as follows. 3 hypotheses are considered: (1) "The task induces a lateralized response", (2) "In novices, orbital activity is more pronounced than lateral activity", (3) "Prefrontal haemodynamic response as monitored with fNIRS is sensitive to surgical expertise". The value of mean task HbO2 - mean baseline HbO2 is considered for the statistical test. No data aggregation is performed. Independent populations that don't follow the normal distribution are investigated. The Mann-Whitney U Test is used to refute all 3 hypotheses. Additionally, for (1) the data is analysed channel- and experience-wise using the Mann-Whitney U Test and Bonferroni correction. The data for (1) is also paired by averaging across subjects and using the Wilcoxon Sign Rank Test. These tests refute (1) too. The refutation of (1) and (3) contradicts previous findings (Leff et al., 2007, 2008). Activation means and standard deviations are calculated. Activation histograms and channel activation heat maps are plotted. They provide some evidence that (1) and (3) could potentially be confirmed.

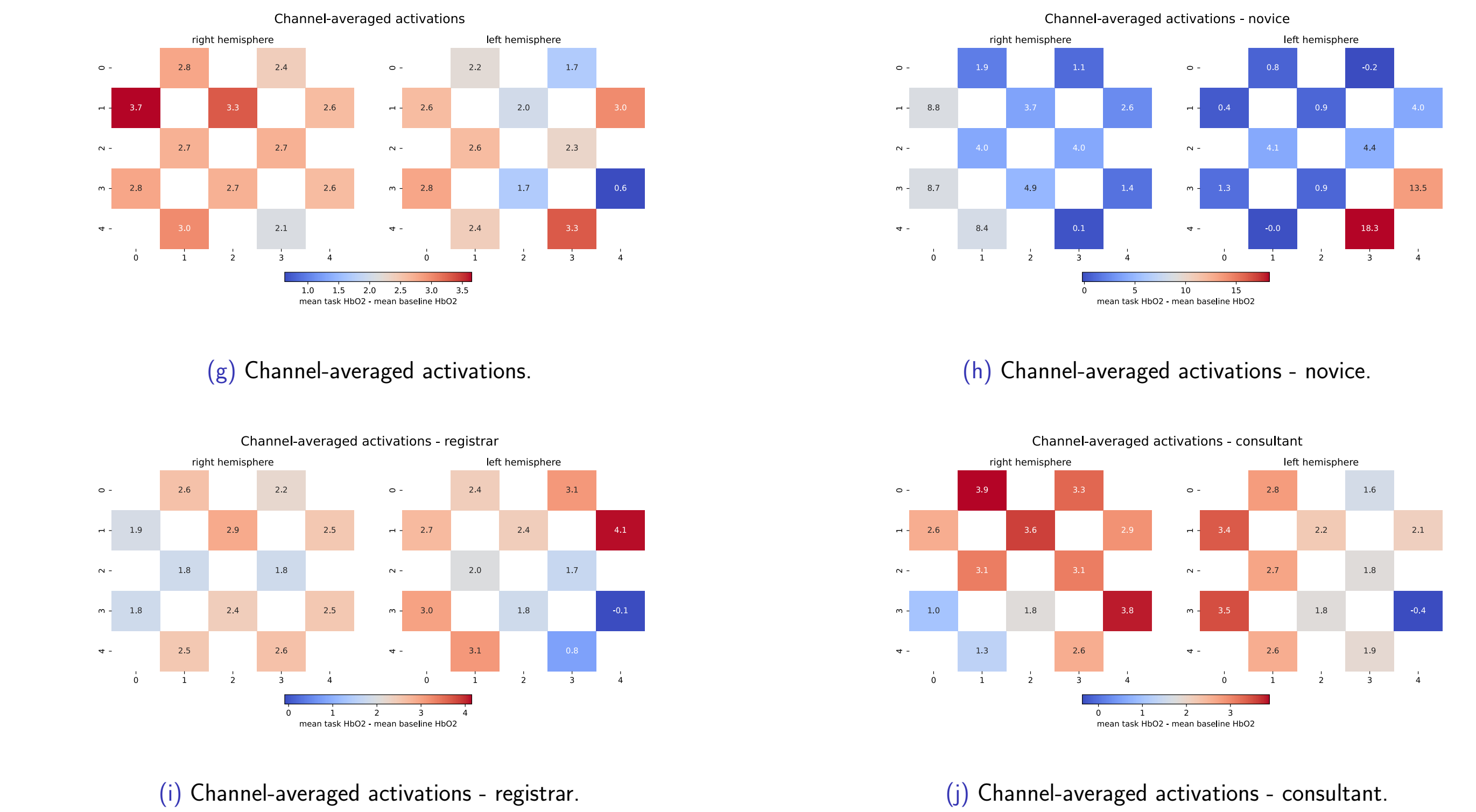
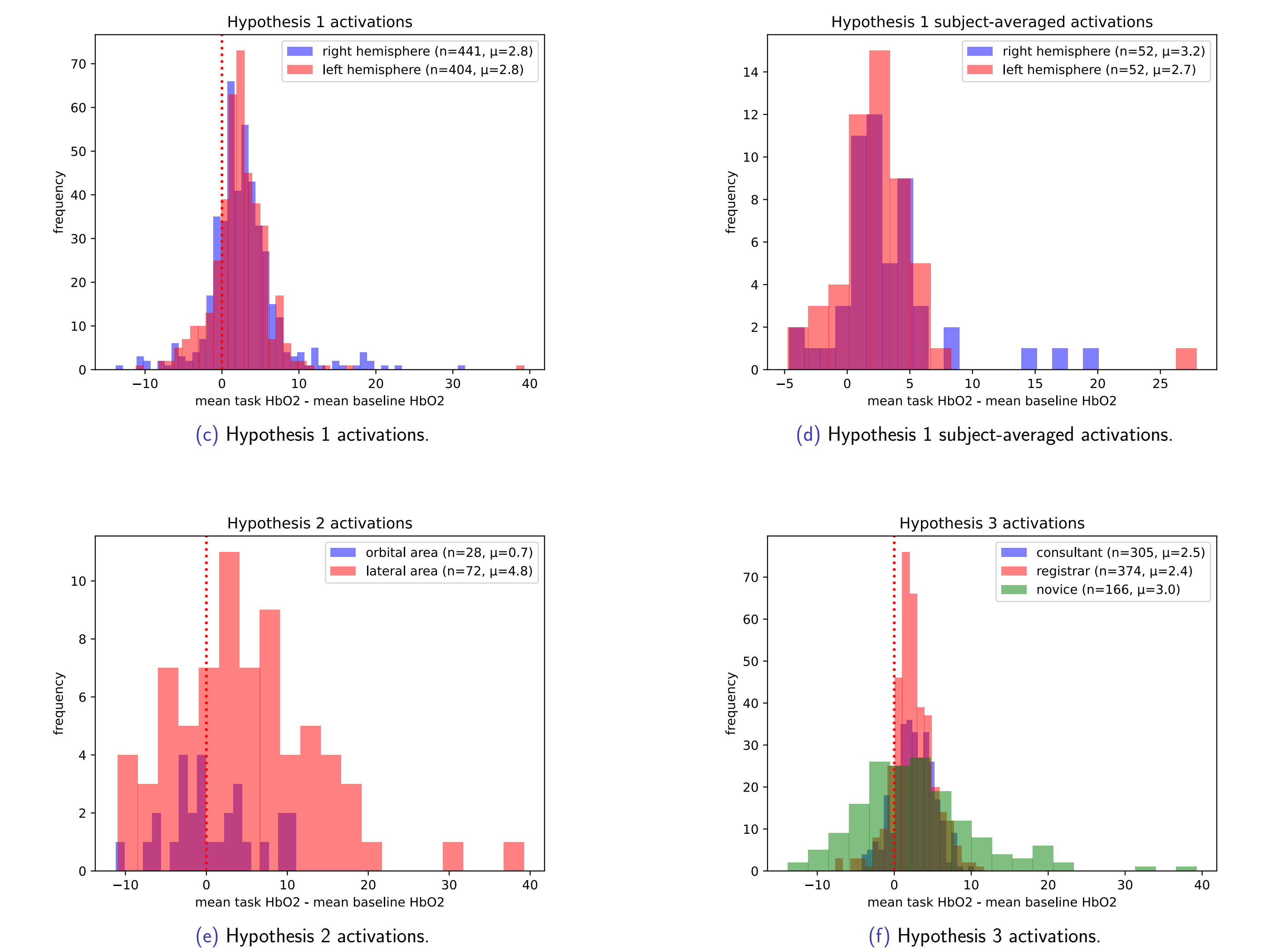


Functional near-infrared spectroscopy (fNIRS) is a non-invasive method of measuring brain activation. It uses near-infrared (NIR) light emitters and detectors to measure changes in volume of oxyhaemoglobin (HbO2) and deoxyhaemoglobin (HHb). Increased HbO2 and decreased HHb levels are often used as a proxy the activation of a brain region, as the region consumes more oxygen and deoxygenated haemoglobin is removed more quickly. One common system for placing the fNIRS optodes on the head of the subject is the Universal International 10/10 system (Jurcak et al., 2007). It assigns points on the head separated by 10% length intervals. The UI 10/10 points used in the experiment are shown in Fig. 1a. The optode channel layout is shown in Fig. 1b. The brain area covered by the optode array is called the pre-frontal cortex (PFC). The most notable regions belonging to the PFC and their corresponding UI 10/10 positions are as follows: Dorsolateral Prefrontal Cortex (DLPFC) - F3, F4; Ventrolateral Prefrontal Cortex (VLPFC) - F7, F8; Orbitofrontal Cortex (OFC) - Fp1, Fp2, F7, F8; Medial Prefrontal Cortex (mPFC) - Fp2, AFz; Ventromedial Prefrontal Cortex (vmPFC) - Fp1, Fp2, Fp2; Frontopolar Cortex (FPC) - Fp2, Fp1, Fp2. The regions investigated in this work (all in hypothesis 2) are: OFC, DLPFC and VLPFC. The experiment investigates surgeons of 3 different skill levels: novice, registrar, consultant. The task involves tying a surgical knot with short (<1 min) breaks in between tying. The existing neuroscience theory allows to make an informed guess regarding the truth values of the 3 investigated hypotheses. Regarding hypothesis 1 (lateralisation), the left PFC is often associated with logical reasoning, sequential tasks, and verbal processes. The right PFC is more linked to spatial reasoning, visual-motor integration, and attention to global patterns. Leff et al. (2007) show that there exists a statistically significant difference in activation patterns between the hemispheres across all expertise levels. Their results show that this phenomenon is the strongest in novices with a stronger left-hemisphere activation compared to right-hemisphere one. Considering hypothesis 2 (orbital activity vs lateral activity), the orbital area (OFC) is activated during tasks requiring decision-making, emotional regulation and evaluating rewards/punishments. Lateral areas (DLPFC, VLPFC) are used for working memory, cognitive control and task planning. Since surgical knot tying involves less emotions and reward/punishment scenarios and more task planning, the lateral areas are more likely to be activated than the orbital areas. Regarding hypothesis 3 (expertise-dependent changes), PFC is activated heavily when learning new tasks, whereas well-known tasks requiring hand-eye coordination cause activation of the pre-motor and sensorimotor cortices (not monitored in the experiment). Hence, it is expected that the overall activation of the PFC decreases with increasing level of expertise. Leff et al. (2007) demonstrate that PFC activation pattern in novices is different than in higher expertise levels. They conclude that PFC activation pattern in registrars and consultants is similar.

The provided processing algorithm works as follows. The Modified Beer Lambert Law is used to extract changes in HbO2 and HHb volume from the recorded light intensities. Baseline correction, downsampling to 1 Hz and detrending is applied. Block averaging is employed. The data is normalised to account for different time lengths taken for different subjects. Channels with faulty data (e.g. due to optode movement) are discarded. The data is averaged across subjects. The following statistics are calculated: haemoglobin volume mean and standard deviation at rest and at task time, mean haemoglobin volume at task time - mean haemoglobin volume at rest time, total volume of haemoglobin that passed through the channel during the task period, time to first peak, time to first nadir after first peak. In hypothesis testing, a wide choice of statistical tests is available for programmatically deciding if a hypothesis should be refuted or confirmed. If the population follows a specific distribution, parametric tests are usually preferred, as they provide a high predictive power. Otherwise, non-parametric tests need to be used. If the data points across the 2 investigated populations are paired, paired tests should be used. Otherwise, independent tests are available. In the case that one wants to determine if 2 populations come from different distributions, a two-tailed test is used. In the case that one wants to show that the mean of one population is higher than the mean of the other population, a one-tailed test is used. Finally, whether the data is continuous, categorical, ordinal or count data influences the choice of the test used too. The most widely used statistic for determining whether the observed phenomenon is statistically significant rather than due to chance is the p-value. If $p < 0.05$, the null hypothesis is rejected. Otherwise, the null hypothesis is retained. To estimate the size of the effect Cohen's d (parametric) and Rank-Biserial Correlation (non-parametric) statistics are often used.

For the statistical hypothesis testing only the mean task HbO2 - mean baseline HbO2 statistic is used. For each population a check is made to determine if it follows a normal distribution via the Shapiro-Wilk test (normal distribution is assumed when p-value returned by Shapiro-Wilk > 0.05). For all hypotheses the Mann-Whitney U test is used as the population datapoints are independent and the populations don't follow any specific distribution. For testing the For hypothesis 1, additionally, the data is averaged across subjects. Since the data is now paired (same subject, 2 hemispheres), the Wilcoxon test is used here. For hypotheses 1 and 3 a two-tailed test is used, whereas for hypothesis 2 a one-tail test is chosen. The analysis is performed using Python with the following libraries: Pandas, Numpy, scipy.stats, matplotlib.pyplot and seaborn. The population pairs compared for the 3 hypotheses are defined as follows. Hypothesis 1: left hemisphere vs right hemisphere. Hypothesis 2: novice OFC (channels: [1, 3, 14, 17]) vs novice DLPFC and VLPFC (channels: [2, 4, 5, 6, 7, 9, 13, 15, 16, 18, 19, 21]). Hypothesis 3: novice vs registrar vs consultant.

Figs. 1c-1f show the histograms of the analysed populations. Fig. 1g shows the overall channel activations during the task. Figs. 1h-1j show the channel activations in different expertise groups.



The calculated p-values for the 3 hypotheses are shown in Table 1. Since all p-values are > 0.05 , all 3 hypotheses are refuted.

test name	p-value	rank-biserial correlation
Hypothesis 1 (fine-grained, Mann-Whitney U)	0.301	0.041
Hypothesis 1 (channel-wise, Bonferonni)	0.985	0.058
Hypothesis 1 (experience-wise, Bonferonni)	0.400	0.070
Hypothesis 1 (subject-averaged, Wilcoxon)	0.985	-0.498
Hypothesis 2 (fine-grained, Mann-Whitney U)	0.985	-0.281
Hypothesis 3 (consultant vs registrar, Mann-Whitney U)	0.218	0.055
Hypothesis 3 (registrar vs novice, Mann-Whitney U)	0.383	0.047
Hypothesis 3 (novice vs consultant, Mann-Whitney U)	0.252	-0.064

Table: Obtained hypothesis p-value and rank-biserial correlation scores.

The obtained results for hypotheses 1 and 3 stand in contradiction to the results obtained by Leff et al. (2007, 2008). Also, the results in the diagrams seem to show that the activation patterns differ between the left and right hemispheres and also between different expertise groups. One reason for this discrepancy might be a poor-quality signal processing pipeline that does not implement bandpass filtering, spatial filtering or optode-movement correction. The present signal processing pipeline uses a one-size-fits-all approach, which might be suboptimal. Another reason could lie in the fact that the statistical pipeline does not classify channels into active and inactive and instead works with noisy continuous values. The refutation of hypothesis 2 agrees with the expected result, as discussed in the introduction.

An fNIRS surgical task brain activity database is analysed. 3 hypotheses are analysed: (1) "The task induces a lateralized response", (2) "In novices, orbital activity is more pronounced than lateral activity", (3) "Prefrontal haemodynamic response as monitored with fNIRS is sensitive to surgical expertise". Using the Mann-Whitney U Test on the value of mean task HbO2 - mean baseline HbO2, all 3 hypotheses are refuted. The refutation of hypotheses (1) and (3) stands in contradiction to previous results. Histograms and heat maps of brain activations are plotted.

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